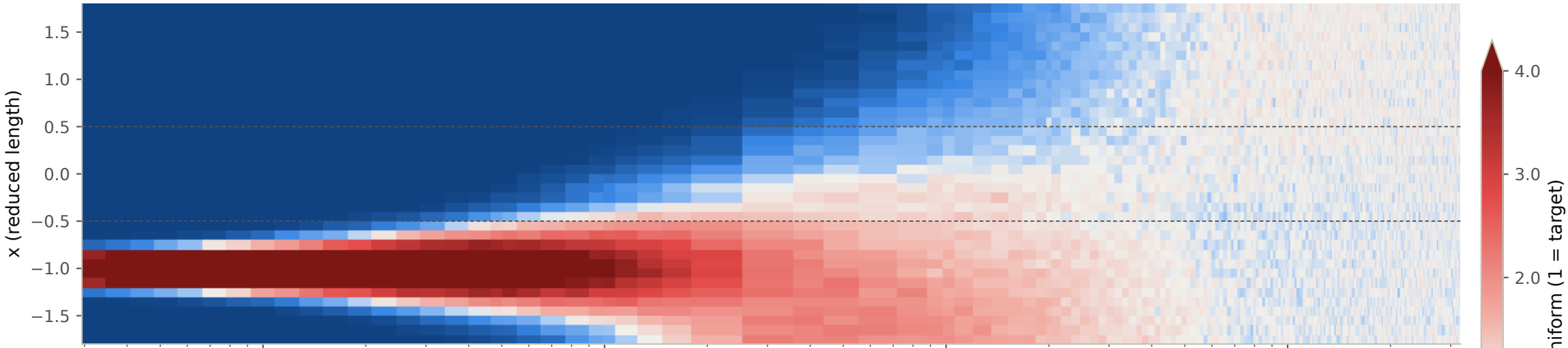


Entropic gateway · $N = 256$ · $T_N = 320$ t.u.

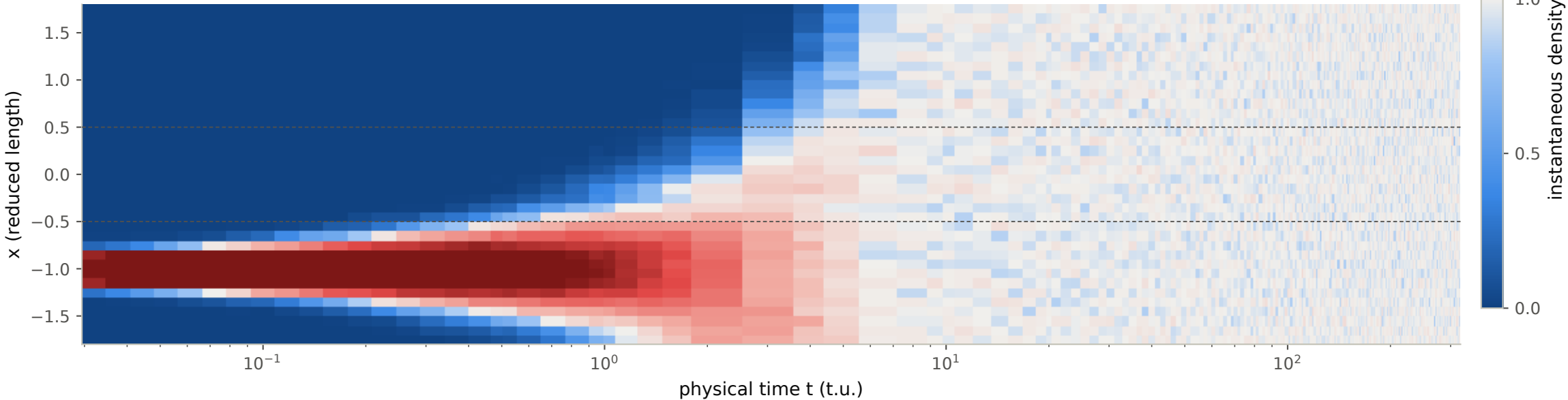
$n_steps = 12,800,000$ · $h = 2.5e-05$ · $B = N n_steps = 3.277e+09$ walker-steps · complete seeds: ABF 32/32, ABF+FR 32/32

Seed-averaged instantaneous CV density vs physical time

(a) ABF (mean over 32 seeds of $hist_inst / N$)



(b) ABF+FR (mean over 32 seeds of $hist_inst / N$)



Identical diverging colour scale for every panel (and every N): blue = below the uniform target, grey = at target, red = above; clipped at 4x. Displayed on 36 CV bins (5 production bins each). Columns = every save (log time; budget grid + physical checkpoints). Dashed horizontal guides: $x = \pm 0.5$ (left well / gate / right well).